Working Paper for Discussion & Review

Title: The Last Mile of AI Ethics:

How AI-Governance Norms Do, and Do Not, Reach the Classroom

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Abstract

The world's principal standard-setters have all converged on a common ethical vocabulary for artificial intelligence. UNESCO's Recommendation, the OECD Principles, and the EU's AI Act now all invoke transparency, human oversight, fairness, and accountability. This paper asks what becomes of that vocabulary during its journey to where AI most intimately touches the young, the school. Drawing on three theoretical lenses - the types of multi-level governance, the international norm life cycle, and the Brussels Effect - and on five expert interviews, it argues that governance norms do not descend a chain of command. They travel laterally, by norm cascade, through intermediaries, and technology vendors. In transit, their meaning is deformed. The safeguard on which every framework ultimately rests - "human oversight" - is also the principle most easily neutered. A human stationed at the end of an automated process, without the information, authority, time, or competence to intervene, satisfies only the letter of oversight while actually hollowing out its substance. Building on Nobel Foundation Strategy Advisor Devashish Sarkar's framework of "value drift through implementation," the paper argues that the decisive governance act is not the drafting of a principle but the ownership of its translation into practice, and where no one owns that translation, the governance of the last mile defaults to whoever remains, which ultimately, is the vendor.

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Introduction

A general-purpose artificial-intelligence assistant arrives inside a classroom. It drafts, summarizes, quizzes, and scores, then it is folded into software a student already opens each morning. Ask that student where the tool came from, and she can name the brand. But ask her what rules govern the tool, who allowed it to read her writing, whether it may be wrong, what recourse she has if it is, and she cannot say. Ask her teacher, her principal, or the school district authority who signed the license, and the answer thins to a shrug. Somewhere far above them all sits a dense and genuinely impressive body of international agreements devoted to answering these questions. But the distance between those agreements and the actual classroom is enormous to cross. This paper is about that distance.

The last several years have produced an unusual convergence in global technology policy. Firstly, UNESCO's Recommendation on the Ethics of Artificial Intelligence, adopted by nearly two hundred states in 2021, has enshrined dignity, transparency, and human oversight as its cornerstones.[1] Secondly, the OECD's AI Principles, the first intergovernmental standard on the subject, had already committed dozens of governments to a near-identical list in 2019, and then revised it further in 2024. Finally, the European Union then translated much of that shared vocabulary into binding law in its Artificial Intelligence Act. Viewed side by side, the three instruments read uncannily like a single global standard just written three times over.[2]

And yet the classroom remains, in any meaningful sense, still ungoverned. This is the critical conundrum the paper investigates. On the one hand, those responsible for policy have constructed an unprecedented density of high-level agreement about how AI should behave, while on the other hand there is a near total absence of any visible governance where it actually behaves. The gap has a name in logistics and telecommunications, the "last mile," the final and disproportionately difficult stretch between a network's trunk and the individual household it is meant to serve. In AI governance the last mile runs from Paris, Geneva, and Brussels to a specific desk in a specific school, and this is where principle either becomes practice or evaporates.

For two reasons, education is the appropriate domain to evaluate how far the gap has stretched. Firstly, this is the precise area that the very tools and technologies themselves identify. In 2023, UNESCO released a formal policy direction toward generative AI in education. The EU considers AI applications within education "high-risk," and ranks them only a category lower than what needs to be totally banned. Secondly, schools usually have the least sophisticated

[1]UNESCO, Recommendation on the Ethics of Artificial Intelligence (Paris: UNESCO, 2021). Adopted by acclamation in November 2021, the Recommendation rests on four core values and a set of principles including transparency and explainability, human oversight and determination, and responsibility and accountability.
[2]OECD, Recommendation of the Council on Artificial Intelligence, OECD/LEGAL/0449 (adopted 2019, amended 2024); Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonized rules on artificial intelligence ("AI Act"), OJ L, 2024/1689.

technology with respect to translating policies into practice. They lack both the staffing capacity to be regulatory bodies and the purchasing ability to acquire complex hardware, software, etc. As such, the thinking goes, if an application can survive the transition from being used by teachers to students in a K-12 setting, it could likely survive nearly any other use case, and conversely, if it cannot survive the educational environment, then the failure is indicative of larger systemic issues.

The research inquiry that gave birth to this paper began in a classroom of a different kind. In a pre-college International Relations program at Tufts University, two bodies of scholarship gave the problem its shape. John Cerone, a scholar of international law, supplied the category. The UNESCO Recommendation and the OECD Principles are instruments of "soft law", normatively weighty, widely subscribed, formally non-binding, and the conundrum of the last mile is, at its core, the conundrum of how soft law comes to bind anyone at all.[3] Enrico Spolaore, an economist of political integration, supplied the counter-weight. The diffusion of innovations across borders is not frictionless but faces long-term barriers, and integration and disintegration often advance together.[4] A norm, like an innovation, does not simply radiate outward from the place it is written. It meets resistance, and is changed by what it passes through.

Holding those two ideas together - that these instruments are soft law, and that diffusion is hard - is where this inquiry starts.

Three sources of evidence support the argument. The first is a close reading of the primary instruments. The second is a trio of theoretical lenses from international-relations scholarship - Liesbet Hooghe and Gary Marks on the types of multi-level governance, Martha Finnemore and Kathryn Sikkink on the life cycle of international norms, and Anu Bradford on the Brussels Effect. The third, and the paper's uniquely distinctive evidentiary base, is a set of five original interviews conducted for this research with experts positioned at different points along the last mile. The first interview was conducted with Devashish Sarkar, Strategy Adviser to the Nobel Foundation, a former CEO of BBC Worldwide Media, and a thought leader on value creation and dilution. The second was with Daniel S. Schiff, a professor of technology policy and co-director of the governance-and-responsible-AI research lab at Purdue University. The third interaction took place with Savneet Singh, who teaches AI ethics at Emory University while also working in technology trust-and-safety. The fourth meeting was with Erin Mote, CEO and Co-Founder of InnovateEDU, a non-profit working on policy in K-12 education. The last

[3]On the category and the problem of definition, see John Cerone, "Soft Law — Stipulating a Definition," in Tracing the Roles of Soft Law in Human Rights, ed. John Cerone (Oxford: Oxford University Press). The author encountered this body of work in the Tufts University pre-college international-relations program; it is cited here as a formative influence rather than as a source for the paper's empirical claims.

[4]See, e.g., Enrico Spolaore and Romain Wacziarg, "Long-Term Barriers to the International Diffusion of Innovations," NBER International Seminar on Macroeconomics 8, no. 1 (2012): 11–46; and Alberto Alesina, Enrico Spolaore, and Romain Wacziarg, "Economic Integration and Political Disintegration," American Economic Review 90, no. 5 (2000): 1276–1296. Cited as a formative influence on the author's framing.

interview was conducted with André Renato Rizzo Dias, faculty member - digital transformation and emerging technology advisor - at the McGill Executive Institute of McGill University, who advises corporations and governments on AI governance and compliance.[5]

One voice among these speaks differently from the rest. Sarkar is not only a leader but also a builder of frameworks for modeling and analysis. His paradigm of how institutions lose their values - "value drift through measurement," "implementation transforms meaning," "competence erosion through automation" - supplies the conceptual architecture of this paper's central sections. The other four experts, from their different vantages, supply the evidence that his mechanisms are not just metaphors but descriptions also of what is happening on the ground. Where canonical philosophy is invoked, it is invoked lightly, as resonance. The theory doing the work here is, for the most part, the theory the practitioners themselves brought to the conversation.

The Convergence at the Summit: What the Instruments Say, and Whether Anyone Hears

Before probing into how norms travel, it is worth determining that there is indeed a shared norm to travel at all. The three principal instruments were drafted by different institutions, for different purposes, with radically different legal force. Their convergence on vocabulary is therefore not a foregone conclusion. It is actually a finding. But a second finding sits beside it, and the two must be held together - the convergence is real at the summit and largely invisible at the base.

UNESCO: Values First, and a Ramp Built for Translation

UNESCO's 2021 Recommendation is the most nearly universal of all the instruments. It is built on four core values. First, respect for human rights and human dignity. Second, the flourishing of the environment and ecosystems. Third, the assurance of diversity and inclusiveness. Fourth, the importance of living in peaceful, just, and interconnected societies. From these, it derives operating principles including transparency and explainability, human oversight and determination, fairness and non-discrimination, and responsibility and accountability.[6]

Two key points stand out. First, the recommendation is honest about its own limitations. It does not presume that values will "happen" of their own accord. Rather, it provides a list of

[5]Devashish Sarkar, interview by the author, 2026; Savneet Singh, interview by the author, 2026; Daniel S. Schiff, written responses to the author's research queries, 2026; Erin Mote, interview by the author, 2026; André Renato Rizzo Dias, interview by the author, 2026. Singh spoke in a personal capacity and not on behalf of any employer. Interview material is treated here as expert testimony rather than as representative survey data; the convergence among independent practitioners is itself part of the argument.

[6]UNESCO, Recommendation, §II (Values) and §III (Principles). The Recommendation is also the first international normative instrument to prohibit the use of AI for social scoring and mass surveillance.

"Policy Action Areas" to assist governments in translating these core values and principles into practice, and education is one area. The fact that a normative document includes a translation ramp from value to policy within the document itself is a quite remarkably early acknowledgment of the precise issue this paper addresses. Second, UNESCO has purposively focused on the classroom. In its 2023 guidelines regarding AI in education, they have taken an unambiguously human-centric stance. The guidelines urge educators and policymakers to protect student's personal data. They also require that students be allowed to maintain control over how the data is being used. Finally, UNESCO's guidelines call for all schools to establish a thirteen-year minimum age for children's use of AI independently.[7]

OECD: A Shared List Whose Own Wording Drifts

OECD Principles, adopted in 2019 and updated in 2024, laid down five values-based principles. The first is inclusive growth, sustainable development and wellbeing. The second is respect for the rule of law, human rights and democratic values, including fairness, privacy etc. The third is transparency and clarity. The fourth is robustness, security, and safety. The fifth is accountability.

The overlap with UNESCO is obvious and intentional. But the 2024 revision is also quietly instructive. Among its changes, the OECD replaced the term "human determination" with "human agency and oversight."[8]

The point is not that the edit is large but rather it is that the summit's own vocabulary is not fixed even at the summit. If "human determination" can become "human agency and oversight" in only five years among the world's most deliberate drafters, one would be justified in expecting far greater slippage as the same words descend to actors with far less time, expertise, and interest in nuance. The drift becomes visible even before the last mile begins.

The EU AI Act: Where Vocabulary Becomes Obligation

The European Union's AI Act is the moment the shared vocabulary hardens into law. Rather than issue principles, the Act sorts AI systems by level of risk. A small set of "unacceptable" uses is prohibited. A larger set of "high-risk" uses is permitted but heavily conditioned. Limited-risk systems carry lighter transparency duties. The remainder is largely unregulated. The AI systems used in education fall into the high-risk tier, and for them the Act demands risk management, data governance, technical documentation, a defined standard of

[7]UNESCO, Guidance for Generative AI in Education and Research, esp. the discussion of a human-centered approach, data-privacy protection, human agency, and the recommended age threshold of thirteen.
[8]Digital Policy Alert, "The 2024 Update to the OECD AI Principles" (2024), summarizing the re-labelling of provisions, including the shift from "human determination" to "human agency and oversight."

transparency, and, in language that will matter a great deal below, that the system be designed so that it can be effectively overseen by natural persons.[9]

Two structural facts recur. First, the Act's central obligations for high-risk educational systems become applicable only as its timeline grows mature. So, the hard-law layer is, at the time of writing, barely beginning to bind. Second, the Act touches any provider placing a system on the EU market, wherever that provider sits. As such, it is an extraterritorial design that is the hinge on which a later argument turns.

Convergence on Words, Divergence on Force - and on Reach

The three instruments share a common vocabulary but they differ in the level of bindingness. It is *recommended* by UNESCO. The OECD *suggests* it. The EU *compels,* but only within its internal marketplace, and only insofar as it develops over time. In Cerone's terminology, two of the three instruments are examples of soft law, and the third instrument is an example of hard law that originated from soft law. The shared words create a powerful impression of a single global standard, but one that their divergent enforceability does not actually support.

In addition, there is another, albeit less apparent, point of divergence. Erin Mote, whose organization tracks the data on the ground, has found that approximately eighty-five percent of students use generative A.I. So do approximately eighty-six percent of teachers. At the same time, however, nearly fifty percent of parents indicate that they have never received any information from their school districts regarding the manner in which A.I. is utilized. [10]

Theorizing the Descent: A Lateral, Non-Hierarchical Model

The intuitive picture of how these norms reach a school is a ladder. The United Nations sets a principle at the top-most rung, a national government adopts it at the rung immediately below, a state operationalizes it at the third rung, and a district implements it finally at the bottom rung. It makes for a convenient picture. But that picture is almost entirely wrong, and understanding why is the analytical heart of the model.

Two Types of Governance (Hooghe and Marks)

Liesbet Hooghe and Gary Marks distinguish two ideal types of multi-level governance. Type I resembles federalism. Authority is dispersed to a limited number of general-purpose, non-overlapping, durable jurisdictions nested tidily one below the other – essentially, the ladder of the intuitive picture. Type II is different in kind. Authority is dispersed to many task-specific, overlapping, flexible jurisdictions that are not arranged in tiers or rungs at all.[11]

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[9]AI Act, Arts. 5 (prohibited practices), 6 and Annex III (high-risk classification), 13 (transparency and provision of information to deployers), 14 (human oversight), and 50 (transparency for certain systems). The core obligations for high-risk systems phase in through 2026; the Act applies to any provider placing a system on the EU market, regardless of where the provider is established.

[10]Mote, interview. The figures Mote cites draw on survey work she discusses in connection with her organization's policy and evidence efforts.

[11]Liesbet Hooghe and Gary Marks, "Unraveling the Central State, but How? Types of Multi-level Governance," American Political Science Review 97, no. 2 (2003): 233–243.

The governance of AI in education is a Type II system. All the relevant actors - the UNESCO, the OECD, the European Commission, national ministries and departments, state task forces, districts, professional associations, and the technology vendors themselves - do not occupy rungs on a single ladder. Instead, they occupy overlapping, task-specific jurisdictions, and a district may be touched by several other actors at once, without any of them commanding the others. The intuitively convenient ladder disappears. There is no chain of command down which a UNESCO principle can climb down to the actual ground level where implementation takes place. If the principle arrives at all, it arrives by some other route.

The Life Cycle of a Norm (Finnemore and Sikkink)

The process by which an international norm spreads can be described in a three-stage life-cycle developed by Martha Finnemore and Kathryn Sikkink. First is emergence (Stage One) where the "entrepreneur" of the norm persuades the initial group of adopters to accept the new norm. Second comes cascade (Stage Two), which occurs when enough other actors embrace the new norm after a threshold has been reached where many follow simply because they wish to maintain legitimacy/reputation, as opposed to believing in the norm itself. Finally, in Stage Three, there is real internalization. Once the norm reaches a certain level of ubiquity, it appears to the actors that the decision to accept the norm was never really one of choice.[12]

This explains a phenomenon that is otherwise hard to account for. A school district technology director who has never read a single word of UNESCO nonetheless “just knows” that an AI tool ought to be transparent and that a human ought to remain in charge. She is not consulting the Recommendation. She is expressing a norm that has actually cascaded past its tipping point and has been internalized into simple professional common sense. The value has travelled, yes, but as a diffused assumption, stripped of the specific content and caveats the drafters had attached with such lofty ideals. Internalization moves the word along, while shedding the fine print, and the shedding is where the later trouble begins.

The Brussels Effect (Bradford)

A second route is neither institutional nor normative, but, in fact, commercial. In Anu Bradford’s exposition of the Brussels Effect, the European Union exports its rules through the market rather than through treaties. Because the EU is a large, affluent market with stringent, non-divisible standards, a multinational firm often finds it cheaper to build a single product that meets EU requirements and then sell it everywhere all across the rest of the world, than to maintain one compliant version for Europe and another for everywhere else. Its customers elsewhere thereby receive EU-grade protections which their own governments never legislated.

[12]Martha Finnemore and Kathryn Sikkink, “International Norm Dynamics and Political Change,” International Organization 52, no. 4 (1998): 887–917.

This is the Brussels Effect de facto. When foreign governments later copy EU rules into their own law, the effect is de jure.[13]

When applied to the classroom, the Brussels Effect finds the transmission mechanism that the ladder could never provide. A global vendor who is building an educational AI product must satisfy the AI Act's high-risk obligations, but after doing so, has little reason to incur the expense of stripping those protections out for students in a jurisdiction that does not require them. Thus, the European statute reaches the non-European classroom, not because any non-European authority commanded it, but because the vendor found compliance cheaper to universalize than to partition. The vendor actually becomes the last-mile courier, and, as later sections of the paper will argue, couriers are not neutral.

The Arrival of the Tool: A Practitioner's Structural Correction

These three lenses explain how a norm might travel, but Erin Mote shares an additional structural fact which explains why the descent so often fails before the transmission has even begun. She draws a sharp distinction between two kinds of technology. An "adoption technology", that enters an institution the way the ladder-model and the procurement officer both assume, that is, through procurement, staged pilot exercises, and centralized risk evaluation disciplines. On the other hand, an "arrival technology" does exactly the opposite. It bypasses institutional gatekeepers and penetrates the classroom, in her words, "from the outside in, through students' and teachers' own consumer habits."[14]

Generative AI, Mote argues, "arrived", it was not "adopted". The significance of this for this inquiry is considerable. Every governance mechanism that the instruments base themselves on - the pilot, the impact assessment, the procurement review - assumes a process of "adoption". But when a technology "arrives" instead, unbeckoned, those mechanisms do not slow it down, because they are simply bypassed. The social-media precedent is the analogy she shares. Faced with an arrival technology, schools "abdicated their responsibility" and "blocked it at the domain level," and declined to build literacy or governance at all. So, in the case of an "arrival technology", "we cannot not govern," and refusing to govern is itself a governance choice, and arguably the one with the worst record.[15]

The model, then, is this. Norms reach the last mile, if they ever reach it, through three overlapping channels - a normative channel (the internalized cascade), a market channel (the Brussels Effect, manifesting itself through the vendors), and a professional channel (the bodies that lay down standards and the associations that convert principle into practitioner guidance). But the very tool they are meant to govern does not wait for any of them. It "arrives" first, from

[13]Anu Bradford, The Brussels Effect: How the European Union Rules the World (New York: Oxford University Press, 2020); and Bradford, "The Brussels Effect," Northwestern University Law Review 107, no. 1 (2012): 1–67.
[14]Mote, interview. Mote offers electricity, the internet, and social media as prior "arrival" technologies, and treats social media as the cautionary precedent.
[15]Mote, interview.

the outside in. Governance is, therefore, always running to try and catch up with a technology that has entered through a door that the governance model did not even know existed.

The remaining sections of this paper questions what happens to the norms during that chase and the answer, repeatedly, is that the norms arrive deformed.

The Counter-Current: Value Drift Through Implementation

If the three channels succeed to some extent in explaining how a norm reaches the last mile, they still do not guarantee its condition on arrival. The central finding of this paper is that the journey deforms the cargo. A principle that leaves the summit in the form of a demanding ethical commitment tends to arrive only as a procedural formality, a nod to the principle's name, but with precious little of its substance.

The most searching analysis of the origins, causes, and nature of the deformation comes not from the academic literature but from one of the thinkers I interviewed, Devashish Sarkar, a Strategy Advisor to the Nobel Foundation. As a former CEO of BBC Worldwide Media, he helmed one of the most trusted institutions in the western world. Sarkar has developed a theory of how institutions lose their values, and it is a cognitive framework that maps onto AI governance with uncomfortable exactness. His paradigm provides the intellectual framework for the subsequent analysis in this section, and the other experts supply the corroboration.

Value Drift Through Measurement

Sarkar's explanation begins with how an *unmeasurable value* is displaced by a *measurable proxy*. An organization, he observes, starts by sincerely declaring its core value to be the upholding of the truth. But it then implements systems that measure and optimize engagement, because engagement is quantifiable, and the truth is not, and organizations by their very nature, "need to measure." "No one ever decides to abandon truth. But "the infrastructure - the measurements, the incentives, the daily decisions - all point in a different direction. Thus, over time, the institution *drifts*. The *stated* value remains unblemished on the mission statement, but it is the *practiced* value that actually drives every decision." He terms this as "value drift through measurement," and points out its most disarming, but dangerous, feature. It "never feels like anything is being corrupted. It actually feels like you are running a successful business, and you start intensifying what is making it successful."[16]

What Sarkar describes through his own prism is a strikingly modern version of a very old problem in moral philosophy - the gap between what the tangible reality actually is, and what it ought to be – which re-emerges here as the gap between what can be counted, and what really matters. His contribution has been to locate the slippage not in a fallacy of logical progression, but in the metrics and the culture that drives an institution to collect the metrics in the first place.

[16]Sarkar, interview.

Sarkar goes on to observe that AI intensifies measurement-driven drift in three ways: 1) by making optimization more granular, 2) by making the drift invisible, and 3) by scaling it globally. "Machine optimization makes what is already measurable ever more granular, so the drift runs ever faster, and hides ever more effectively."[17]

The other practitioners also confirm this mechanism from their own respective areas of experience. Daniel Schiff is Associate Professor of Technology Policy at Purdue University's Department of Political Science and the Co-Director of GRAIL, the Governance and Responsible AI Lab. According to him, ethical trade-offs are on several occasions explicitly metricized, that firms "literally have budgets for how much attention or efficiency gains they are willing to sacrifice" to reduce the quantum of harm, and that it took "two-plus decades of social media concerns" before any significant opposition arrived.[18]

Savneet Singh teaches AI ethics at Emory University while also working in technology trust-and-safety. She explains that measurement takes place in layers. The operational metrics - precision, recall, response time, consistency - are the easiest. When it comes to harm severity, it gets harder, and questions of justice and access are harder still. The final layer, whether a system supports trust, agency, and well-being, is "very difficult to measure indeed." This final layer is, of course, exactly where the summit's most important values reside, and it is the first layer to be neglected, because it resists being measured by a number. The consequence for education is direct and profoundly disturbing. A tool which has been procured under the banner of "human-centered learning", when put into operation, optimizes only what it can count - time on task, completion, engagement - but not the human formation that the banner had promised. The value remains unrepudiated. But it gets out-measured.[19]

Implementation Transforming Meaning

Sarkar's second mechanism is even more subtle. The principle survives as structure, but all the while, its meaning quietly changes. At BBC Worldwide Media, the institution held up a principle of "editorial independence", that editorial decisions would be made on news judgement, not business pressure, and then committees, guidelines, and workflows were created to safeguard the principle. But, over time, he explains, the principle "got transformed into only following the editorial processes. People stopped asking 'Is this the right story?' and started asking instead 'Are we following the correct process?'" The value did not vanish. But it was gradually displaced by adhering to the very procedure built to protect it. He calls this

[17]During my discussion with Sarkar, which lasted for several hours, we found resonance with David Hume's observation that claims about what ought to be cannot be derived from claims about what is, as well as with the deeper terrain of the Upanishads of ancient Indian thought, and the difference therein between "namarupa" and "satya". Sarkar's "value drift through measurement" is, in effect, that same difference, only operationalized, with what is measurable quietly substituting for what is valuable.

[18]Schiff, written responses. Schiff cautions that ethical language should be tested against "what actually changes" - what is measured, who is consulted, and whether efficiency is ever sacrificed when human costs are high.

[19]Singh, interview.

"'implementation transforming meaning, and it's one of the most subtle and insidious forms of value drift."[20]

Sarkar then applies the mechanism directly to the governance vocabulary. "Transparency," he analyzes, "which is the demand of UNESCO, literally cannot be delivered for a model that has billions of parameters. So, in the execution journey, *transparency* degrades into *disclosure*." Its meaning alters, in his words, "from 'you can understand how this works' to 'we've told you something about how this works.'" And "human agency," the principle every one of the guiding frameworks invokes, also *degrades* along an identical path. It starts meaning "you can click to override if you want," but afterwards, "over time, people simply stop overriding because the AI is usually right. So, the principle is still there, surviving in the structure, but it is being continuously hollowed out in practice."[21]

The paper follows this insight into its central section. It is also a point on which the other experts wholeheartedly agree. But before turning there, two further mechanisms need to be discussed in order to completely incorporate Sarkar's paradigm.

The Invisibility of Embedded Values

Sarkar's third mechanism concerns values that never even surface, thus rendering them impossible to debate. For most of history, he explains, authority was visible, and the visibility made it contestable. A textbook, a teacher, an institution, could be questioned. But an AI system's output arrives instead as appearing to be perfectly neutral information, but all the while, the value-laden choices which shaped it - what data to train it on, whose language to prioritize and privilege, and even more ominously, what to suppress and deprioritize - remain deeply interred inside code and datasets. Sarkar shares a deeply unsettling analogy. The filmmaker Leni Riefenstahl "may not have been consciously trying to help Hitler… But her art form itself - cinema, with its powerfully visceral impact - carried values," invisibly, embedded inside the very grammar of the medium. Modern AI, similarly, "is value-rich by its very design," but "all those values remain invisible."[22]

This is Sarkar conceptualizing squarely within the tradition of media theory that he invokes elsewhere also. This line of thought, associated with Marshall McLuhan, holds that the medium itself is a carrier of meaning, one that is irreducible to its content alone. But Sarkar extends this phenomenon to the AI training set and the model, and he points out that an AI system is a medium whose embedded values are far less visible, and therefore far less contestable, than those of any newspaper or TV channel. This puts the governance instruments

[20]Sarkar, interview.
[21]Sarkar, interview.
[22]Sarkar, interview. The Riefenstahl analogy is Sarkar's. He discusses the work of the famous film maker who wanted to create new frontiers in the medium, and pioneered many techniques that are still widely used, but whose reputation suffered because of the positive image she created for the Nazis. He argues that a medium can carry and propagate values independently of any conscious intent on the part of its creators.

under severe strain. When it comes to the UNESCO demands of transparency and explainability, Sarkar feels that the demand may be unmeetable at the level the principle originally imagined, so "*transparency*" dilutes to *disclosure* and the embedded choices persist, in his phrase, as a case of "outsourcing epistemology to systems we don't understand, that are operated by companies we can't trust, and in service of values we can't see."[23]

Competence Erosion

Sarkar's fourth mechanism is the most recursive, because it actually attacks the very same human capacity the other principles presuppose. When algorithms begin making the judgments that journalists once used to make - what is newsworthy, what is credible - journalists "stop practicing judgment, and start adapting to the algorithm instead." The skill atrophies, he says, "not because people are not working hard, but because the system doesn't require the skill anymore. And then the old phenomenon - use it or lose it - takes over." He finds it cruelly ironic that the younger practitioners are the ones "more competent with the tools…but less competent in the fundamental human skill" that *the tools were meant to serve*. He describes this as "competence erosion through automation."[24]

Singh identifies the classroom version of this phenomenon. Her most worrying concern about students is the over-reliance - the effect of leaning, ever increasingly, on AI companions and, even more dangerously, the deadly serious "intellectual harm that results from mistaking output that flows out fluently for output that is the truth." Thus, the student who outsources the labor of reasoning to a fluent machine starts losing the capacity to distinguish the truth, and thus losing the very capacity that oversight of the machine would require. Mote frames this same loss insightfully as a matter of where learning itself starts to crumble. "All learning takes place while writing the first draft," she explains, "so if you're using AI to do your first draft itself, you'll eventually stop learning."[25]

So, it is the recursion that is the point, and it now sets up the section that follows. Every governance instrument that has been adopted leans on human oversight as the ultimate safeguard, but when oversight presupposes a competent overseer, so the same automation the instruments are attempting to govern is actually eroding the competence on which their central safeguard depends. The principle keeps assuming the capacity exists, while it is actually being destroyed.

[23]Sarkar, interview; and, on the medium as message, the lineage of Marshall McLuhan, Understanding Media: The Extensions of Man (New York: McGraw-Hill, 1964). Sarkar invokes McLuhan explicitly in the interview.
[24]Sarkar, interview.
[25]Singh, interview; Mote, interview.

Drift Is Not Corruption

But none of this is mala fide. One of this paper's sub-arguments is that value drift is not the result of conscious efforts made by bad actors. Instead, it is a by-product of ordinary organizational life. According to Schiff, the gap between what is committed and what is practiced is "not necessarily that companies say one thing and secretly do another." More often, it is the case that broad commitments "get watered down as they move through processes, and teams strive to meet product deadlines, lawyers give their inputs, engineering faces constraints, and overall ownership is unclear," and the hardest questions, like who is responsible when a system causes harm, and "whether anyone has the authority to slow down, or change course", start going unanswered. "The drift often happens in everyday organizational practice, and not only in public messaging."[26]

The Hollowing of Human Oversight

All the governance frameworks surveyed in this paper come to rest, in the end, on that one single safeguard. When technology cannot guarantee the principle, the human being will. There will be a natural person who will supervise the system, detect its errors, and, if necessary, overrule it. Human oversight and determination are a core principle for UNESCO. The OECD elevated "human agency and oversight" in 2024. The EU requires that high-risk systems be designed in a manner that allows them to be "effectively overseen by natural persons." So, human oversight is the load-bearing column of this entire edifice. And, based on the convergent testimony of all five experts interviewed for this paper, who never spoke to one another, the column most likely to be completely hollow, and a human placed at the end of an automated process does not, just by that placement, retain control.[27]

Sarkar provides the theoretical underpinnings. The principle is abstract. Therefore, it needs to be embodied in a concrete mechanism, and that mechanism - an override button, or a sign-off step - then evolves into a proxy that the system starts optimizing around. [28]

Singh, Dias, Mote, and Schiff demonstrate how this actually happens.

Singh: Placement Is Not Agency

Singh, who researches this provision first hand, argues that the field has rubbed the phrase smooth. "We overuse the term human in the loop," she says, and the vagueness conceals some real and important distinctions - "human over the loop, human out of the loop, and AI in the loop, all of those terms make a lot of difference." Her central warning is the one the governance instruments most need to hear. "By simply placing a human being… at the end of an

[26]Schiff, written responses.
[27]UNESCO, Recommendation, principle of human oversight and determination; OECD, Recommendation, §1; AI Act, Art. 14.
[28]Sarkar, interview.

automated process does not… preserve agency," because that person "might lack the context, and may not have the time or the confidence needed to make that decision. So, 'human in the loop' just… fails." Singh points out that agency is not mere presence at the end of a pipeline, but it is the real capacity to choose, and make decisions such as - "when to override… when to consent… when to just shut it down."[29]

Dias: The Three Conditions, or It Is a Fiction

André Renato Rizzo Dias, who advises corporations on exactly this problem, converts Singh's critique into a specification. For human oversight to be real rather than ceremonial, he argues, the overseer needs three things: 1) information, 2) authority, and 3) the time needed to act. The overseer needs information to actually understand what the system is doing. She needs authority, in order to act on that understanding. "You must have the power to override the product, to switch it off, if necessary", without being overruled or. And, most importantly, "you need the time to act, and this is what AI threatens the most."[30]

He explains that with the rise of autonomous agents, "you have hundreds, sometimes thousands of agents taking actions… 24/7," and "no human can ever keep track." An overseer who cannot keep pace, cannot oversee, no matter how much information and authority is granted to her. Human oversight, Dias concludes, is effective only when all three conditions are met, and even if any one of them is missing, you have what he calls "governance fiction", where there is a structure that "lets everyone feel safe while nothing at all is actually being overseen."[31]

Dias also shares the mechanism by which oversight quietly lapses, and it rhymes exactly with Sarkar's notion of drift. Authority migrates from human to machine not by decision but out of habit. A system first offers information, which then becomes a recommendation, which, proving useful, then comes to be accepted as a decision, and after some time elapses, "the culture of the organization changes, and it becomes one where there is no questioning anymore… information morphed into recommendation that morphed into decision, because everybody trusted, and nobody questioned it." And "the moment they realize this is happening, most of the time it's already too late." Oversight is never revoked. But it is allowed to atrophy, one unquestioned recommendation at a time.[32]

Mote and Schiff: The Overseer Who Cannot

Mote calls out that you reach the end state of this drift when you are not merely delegating a task but also surrendering your judgement along with it. She distinguishes ordinary "cognitive offloading" such as relying on a map application for directions, from "cognitive

[29]Singh, interview.
[30]Dias, interview.
[31]Dias, interview.
[32]Dias, interview. Dias characterizes the shift from recommendation to decision as "more like a drift" than a conscious delegation of authority.

surrender," when you start relying on the system to do your thinking, and "actually substituting for your own judgment." When it comes to oversight, this leaves you with an overseer who has surrendered her underlying judgement, and becomes an overseer in name only. Her summary of what still remains with the human is the section's moral. "AI is making knowledge ubiquitous, but it lacks wisdom. And so, what do humans bring? They bring wisdom and judgment." But a framework that allows the human to only sit in the judgment seat, while permitting the surrender of the judgement itself, preserves nothing.[33]

Schiff explains how he tests whether the organization's "human-centered" language is real by asking among other things, "whether people can challenge decisions" and "whether the organization is willing to give up some efficiency when the human costs are too high." Those are also precisely Dias's "authority" and "time needed", but stated as organizational will rather than individual capacity. Drift, for Schiff, are commitments watered down through "unclear ownership" and the unanswered question of "whether anyone has the authority to slow down the system." These are how institutions hollow out the overseer.[34]

Five Experts, One Conclusion

The convergence is the argument. Sarkar derives the hollowing from the structure of implementation. Singh demonstrates that placement is not agency. Dias identifies the three conditions whose absence makes oversight a fiction, and names the drift that erodes them. Mote points to the surrender of judgement that empties the overseer from within. Schiff locates the organizational failure that leaves the overseer unempowered. The philosophical stakes are now explicit, and they were called out in the interviews themselves. An issue in "human oversight" is human agency, the capacity, as one interviewer put it, to be "the author of your own life." When a governance framework secures the appearance of oversight while permitting the erosion of agency, it not only fails a technical requirement, but it also permits a quiet alienation of the person from their own judgement, and then goes on to call that compliance.[35]

Who Governs the Last Mile? Ownership, Procurement, and the Vendor

If the drafting of principles, and their internalizations as diffuse assumptions do not govern the classroom, and if the safeguard of last resort is so easily hollowed, what actually governs the last mile? The interviews converge on an answer the governance instruments hardly mention. The last mile is governed, if at all, at the point of procurement, by whoever owns the translation of principle into product, and, by default, that owner is actually the vendor.

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[33]Mote, interview. A "human in the loop who reviews a thousand decisions an hour with no time to evaluate does not meet the point of the rule."

[34]Schiff, written responses.

[35]The phrase arose in the author's interview with Mote, in an exchange about whether "cognitive surrender" should be understood as a form of alienation, a separation of the person from the very capacity that lets them author their own life. Mote's response, that the purpose of education is "not schooling, it's learning… fundamentally, to gain wisdom," treats the preservation of that capacity as the point of the whole enterprise.

The Vendor as Effective Regulator

The Brussels-Effect argument established that the vendor is the courier that carries a European statute into a non-European classroom. Sarkar draws out what the courier metaphor conceals. The vendor delivers not only protections but values as well. The capacity to determine what information reaches people is now, he argues, concentrated in a handful of firms accountable "not to the communities they serve, but to shareholders," whose systems "embed their values… about what matters, what's true, what's acceptable". The values are "not neutral". They are "the values of Silicon Valley engineers… deployed globally." A district that inherits an educational AI through the vendor channel inherits, in that same transaction, both the protections it never negotiated, and the value choices it never debated.[36]

For Dias, it's the same inheritance, only from the buyer's side. The organization buying a vendor's AI system is "going to pay for a software that was developed by engineers that you never met before, who have no idea about your organization." The system's values and biases arrive pre-installed. Yet, the accountability does not travel with them: "you can outsource the algorithms… the development… But the accountability for the outcomes… it still has to be you. It's your brand." This is governance by a party that stood for no election and drafted no principle, that will be exercised over a buyer who retains all the liability, but has ceded all the control.[37]

Sarkar adds a qualification that further complicates the Brussels-Effect story. Even where the European standard being exported is genuinely protective, it is not universal. It only "reflects a local history, a local philosophy, a local culture," and when applied globally, it can suppress content that may be "perfectly accurate and important in the local context", because it triggers a rule "designed for Europe." The vendor, then, is no longer just delivering protection unevenly. They are thrusting one civilization's values down onto classrooms that never consented to them.[38]

Procurement as the Real Site of Governance

The vendor governs by default, so the only moment at which anyone else can govern is at the time of purchase. This is Schiff's central recommendation for practical implementation. School leaders, teachers, and students, "need to understand AI governance through concrete cases, and not rely only on abstract principles." The real points of evaluation are not which values to recite but "who built it, who bought it, what data does it use, what does the contract allow, who reviews it, and what happens if it harms students or teachers?" and, then very pointedly, "was there ever a randomized evaluation done or a pilot study, or are these just shiny vendor claims?" This is the kind of "governance literacy," he contends, that "makes power visible," and which "so often is the missing piece."[39]

[36]Sarkar, interview.
[37]Dias, interview.
[38]Sarkar, interview.
[39]Schiff, written responses.

Mote sharpens this further. A tool that is "FERPA and COPPA *aligned*" may not be *compliant* at all, and the phrasing "obfuscates the truth." But her solution to the problem, a short rubric of quality indicators, carries its own hazard, one she recognizes herself. The rubric itself can ossify into "a checklist", and then it becomes the very implementation-transforms-meaning drift Sarkar describes, but now reappearing in the very tool meant to prevent it.[40]

The Ownership Thesis

How, then, does a principle survive? Singh says that "principles and values survive implementation when someone owns the translation and, when the systems affected by them have the ability to test it out." The key and final resource is not principle, not product, but the person who sits astride both, who "can see both the worlds," who can articulate that "transparency for this product means this" and that "a bias-free product looks like this."[41]

Singh very insightfully observes that the translation gap arises because the organizations simply don't have the skills, a "vacancy." Dias describes the same vacancy as caused by culture not yet suited to the task. "People need to be able to say that there is something wrong." Where the vacancy remains unfilled and the culture has not been built, the vendor's defaults will continue to govern. Only when someone owns the translation, and is actually informed and empowered to act on it, only then does the principle have a chance.[42]

The Asymmetry of Voice, and the No-Sovereign Problem

There are two other gaps. The multi-stakeholder principle of UNESCO insists that those who are affected by AI, mainly children, must have a say, but Schiff describes how practice falls short. At international and federal levels "experts, standards bodies, etc., tend to dominate," but in K–12, students, who are probably the most affected group, usually have the least power." This is a structural "asymmetry between who is the one affected and who is the one who has power." He proposes modest and testable solutions, such as student groups, gathering feedback during pilots, and impact assessments.[43]

The second gap lies in the lack of a sovereign authority to resolve disputes. Dias, who advises across borders, counts "eighty-four new AI-related laws in twenty-seven different states" in the first half of 2026 alone, and warns that a single global authority is simply not possible in the short or medium term. Governance will instead work in layers, with laws, industry standards, private contracts, all operating simultaneously, and the situation will require a new competence he terms as "tech diplomacy." His deepest reframing looks at trust itself. "The question is not if

[40]Mote, interview. The quality indicators she proposes - safe, interoperable, usable, inclusive, evidence-based - are offered as a diagnostic rubric, with the explicit caveat that a rubric can decay into box-ticking.
[41]Singh, interview.
[42]Singh, interview; Dias, interview.
[43]Schiff, written responses.

we trust AI or not. “It’s about whether we are going to trust the organizations that are using AI to make such a decision. Should I trust these organizations who are using AI on me - as a citizen, as a student? This is a more philosophical question.” The governance of AI is gradually emerging as a question about the legitimacy of delegated authority, and about whether the institutions who wield these systems over us have actually earned the right to do so.[44]

Complication: The Withdrawing Middle and the Limits of the Argument

We must also now confront a development that appears to break the model. Between the drafting of the governance instruments and the writing of this paper, one government has actually reversed course. In the United States, the 2023 executive order on “Safe, Secure, and Trustworthy” AI was revoked in January 2025 and replaced by an order that will remove regulatory barriers, in order to sustain “America’s global AI dominance.” The technical guidance that was mandated earlier has now become only voluntary.[45]

But when a governmental tier withdraws, this doesn’t stop the transmission, it only changes the direction. The normative or professional expectations persist. There is an implicit expectation from professionals as well as vendors that they will continue to remain transparent, because of their professional responsibilities. That obligation does not end just because an executive order was rescinded. On the contrary, the market channel becomes stronger, relatively speaking. Vendors who have built to the standards of high risk under the EU's AI Act carry these obligations forward, be it in the U.S. or Europe, or indeed, anywhere else. But the transmission now becomes even more dependent upon the least accountable of the remaining tiers - the vendor and the professional intermediary. As such, in the U.S., we are left with what can be called "governance-by-the-remainder," a new regime driven by whoever has not retreated, which Mote and Dias both indicate is the Vendor.

There is a countervailing note worth recording, because it complicates the picture in the other direction. Mote observes that sub-national and foreign actors have, in several instances, moved decisively where the federal center did not. In her own state of Florida, there has been swift action to remove certain AI companion products from classrooms. Outside the US, there have been national measures against anthropomorphic consumer chatbots. Governance-by-remainder, in other words, is not only governance-by-vendor, it is also governance by whichever public actor retains both the will and the authority to act. The lateral model does not predict the collapse of governance so much as it predicts its unpredictable redistribution.

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[44]Dias, interview.

[45]Executive Order 14110, “Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence” (October 30, 2023), revoked by Executive Order 14148 (January 20, 2025); Executive Order 14179, “Removing Barriers to American Leadership in Artificial Intelligence” (January 23, 2025). The NIST AI Risk Management Framework (2023) remains in use as a voluntary framework.

The limitations of the arguments of Dias and his allies need to be identified as clearly as do their claims. This paper focuses solely on one area - education. No evidence is offered for the notion that the same structural conditions exist in the same manner across other sectors. In addition, while Dias's testimony concerning corporate interests supports the idea that the structure is not limited to education, the evidence used to support Dias' position is based upon expert opinion, not generalizable data. Therefore, while Dias provides insight into the mechanism and demonstrates their alignment, he does not provide, yet, an assessment of their frequency or extent. Additionally, since the vendor-transmission pathway is well-supported both theoretically, and through the views of the experts who were interviewed for this paper, this pathway has been inferred, and not specifically demonstrated using contract-level data. Document-based evidence showing how EU obligations are implemented in practice would have significantly enhanced the strength of this inference. Finally, the analysis presented in this paper is primarily focused upon the relationships among UNESCO-OECD-EU-US. It thus minimizes consideration of the Global South, as pointed out by Sarkar, and therefore represents a cautionary note that exported European values are not universally applicable.

Conclusion

The summit has converged. Within a few years UNESCO, the OECD, and the European Union arrived at a shared ethical vocabulary for artificial intelligence - transparency, human oversight, fairness, accountability - and the European Union has begun to give that vocabulary the force of law. The last mile has not converged, and has scarcely been governed. Between the principle and the classroom lies a Type II thicket down which no command can travel. The norms that do arrive do so by cascade, by vendor, and by intermediary, while the tool they are meant to govern arrives first of all, from the outside in. And in the passage, their meaning drifts. Transparency arrives as disclosure. Human agency arrives as a button rarely pressed. Human oversight - the safeguard on which all the others depend - arrives as a person stationed, without information, authority, time, or the judgement long since surrendered, at the end of a process they cannot actually stop. The value survives as structure and thins in substance, not because anyone corrupts it, but because no one owns the labor of keeping it whole.

The underlying issue is that the governance instrument encourages the opposite of the practical implications. When a value does not reach practice, the first reaction is to repeat it louder within a new framework. However, if the reasoning presented in this work is correct, then repeating the value will also be incorrect. The problem is clearly a matter of translation, not of declaration. What the final mile needs is not additional principle, but owned and testable practice. An impact assessment that occurs prior to a procurement decision, a reading of contracts to determine what they allow, and an auditable record of how a system makes decisions. Human oversight should have Dias's three conditions available to them, as opposed to simply pressing a button. There should exist real, non-performance- based channels through

which the students impacted by the systems are able to communicate. Most importantly, institutions need to measure the values they claim to uphold, instead of measuring their ability to use proxies for the values. [46]

None of the practitioners interviewed for this paper is naive about how hard this is, and neither is its author. Sarkar predicts that it is the generation now in middle and high school, having been harmed most by the absence of trustworthy information, that will prove to be those most "fierce" in rebuilding it, and will seek "solutions that change the technology interface itself." Mote reminds us that the purpose of the whole enterprise lies in what the machine lacks. Education is "not schooling, it's learning… fundamentally, it is about gaining wisdom." Singh ends in a role rather than a rule. The owner of the translation must be someone who can see both worlds and refuses to let a value dissolve into a checklist.

Between the expectation and the role lies the real work of AI governance. It is not done at the summit, where the words have already been written. It is done, or left undone, in the last mile, in the purchase order, the pilot, the override that is or is not genuinely available to a thirteen-year-old girl, by whoever is willing to own what a principle actually requires, once it stops being a principle, and reveals itself to be actually a machine inside a classroom computer that is quietly studying the child constantly.[47]

[46]Singh, interview. The prescriptive turn synthesizes the interview testimony: Schiff's emphasis on procurement, evaluation, and governance literacy; Dias's three conditions for genuine oversight; Mote's insistence on evidence and on measuring the right things; Sarkar's demand that institutions continually ask whether their practice still matches their principles; and Singh's ownership of the translation.
[47]Sarkar, interview; Mote, interview; Singh, interview.

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